HDOC Day-57

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Batch	15

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **arr[]**. Pass the array to a function named **max_subarray_sum()** using its address. Use pointers to pass the array between the calling function and **max_subarray_sum()**.

Find and display the maximum sum of any contiguous subarray using **Kadane's algorithm**. If all elements are negative, the result must be the largest (least negative) element. The selected subarray must contain at least one element.

C Program

```
#include <stdio.h>

void max_subarray_sum(int *arr, int size)
{
    int i;
    int currentSum = *arr;
    int maxSum = *arr;

    for (i = 1; i < size; i++)
    {
        if (currentSum + *(arr + i) > *(arr + i))
            currentSum = currentSum + *(arr + i);
        else
            currentSum = *(arr + i);

        if (currentSum > maxSum)
            maxSum = currentSum;
    }

    printf("Maximum subarray sum: %d\n", maxSum);
}

int main()
{
    int size, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", arr + i);

    max_subarray_sum(arr, size);

    return 0;
}
```

Explanation

The array address is passed to **max_subarray_sum()** through a pointer. Kadane's algorithm keeps two values: **currentSum**, the best sum ending at the current position, and **maxSum**, the best sum found so far. At each element, the program decides whether to extend the existing subarray or start a new one. Both variables are initialized with the first array element, so an array containing only negative values is

handled correctly.

Sample Outputs

```
Example 1
Array: -2 1 -3 4 -1 2 1 -5 4
Maximum subarray sum: 6

Example 2
Array: -8 -3 -6 -2 -5 -4
Maximum subarray sum: -2

Example 3
Array: 5 -2 3 4
Maximum subarray sum: 10
```

Result

Thus, the program successfully finds the maximum contiguous subarray sum using pointers and Kadane's algorithm, including the case where all array elements are negative.